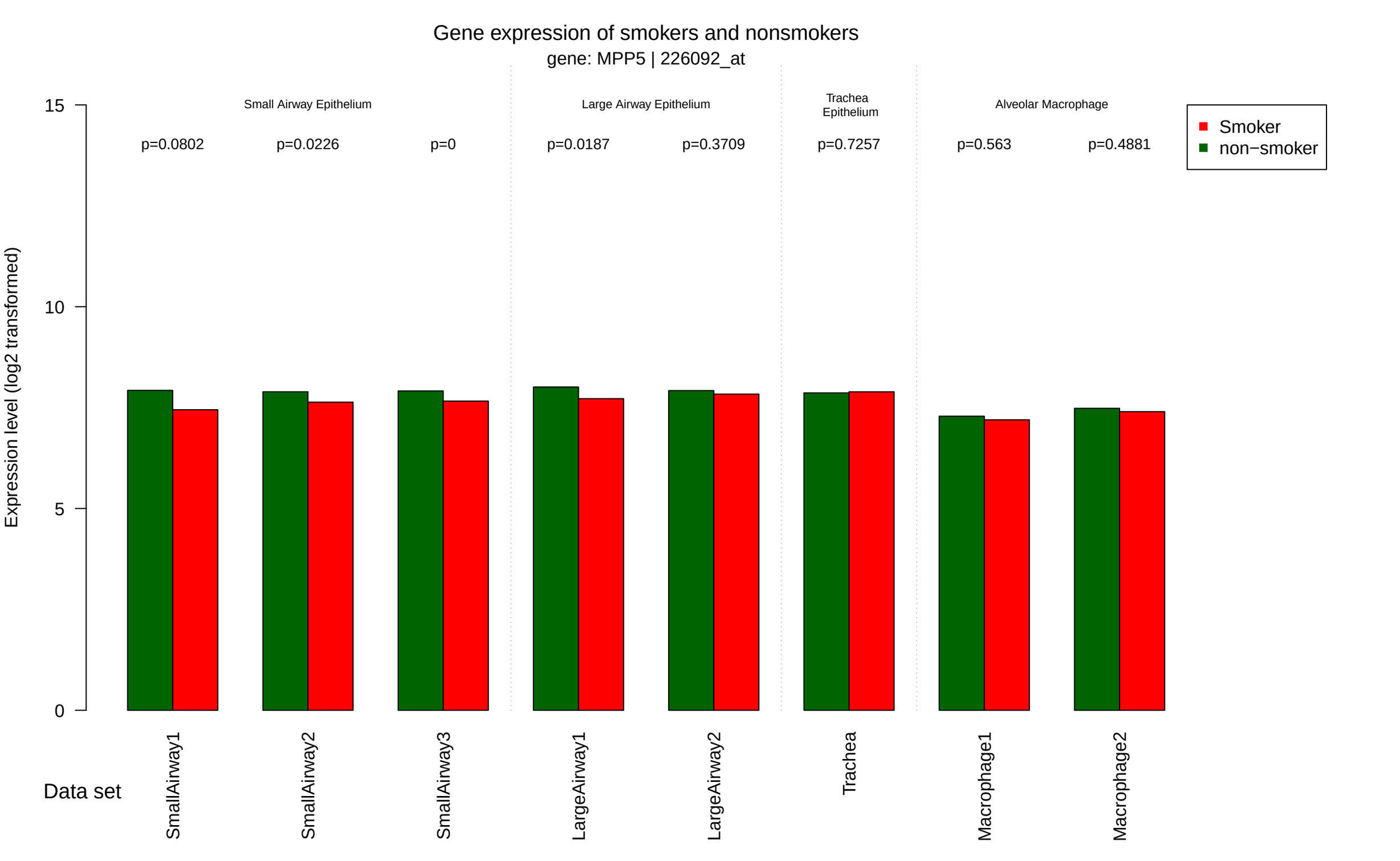
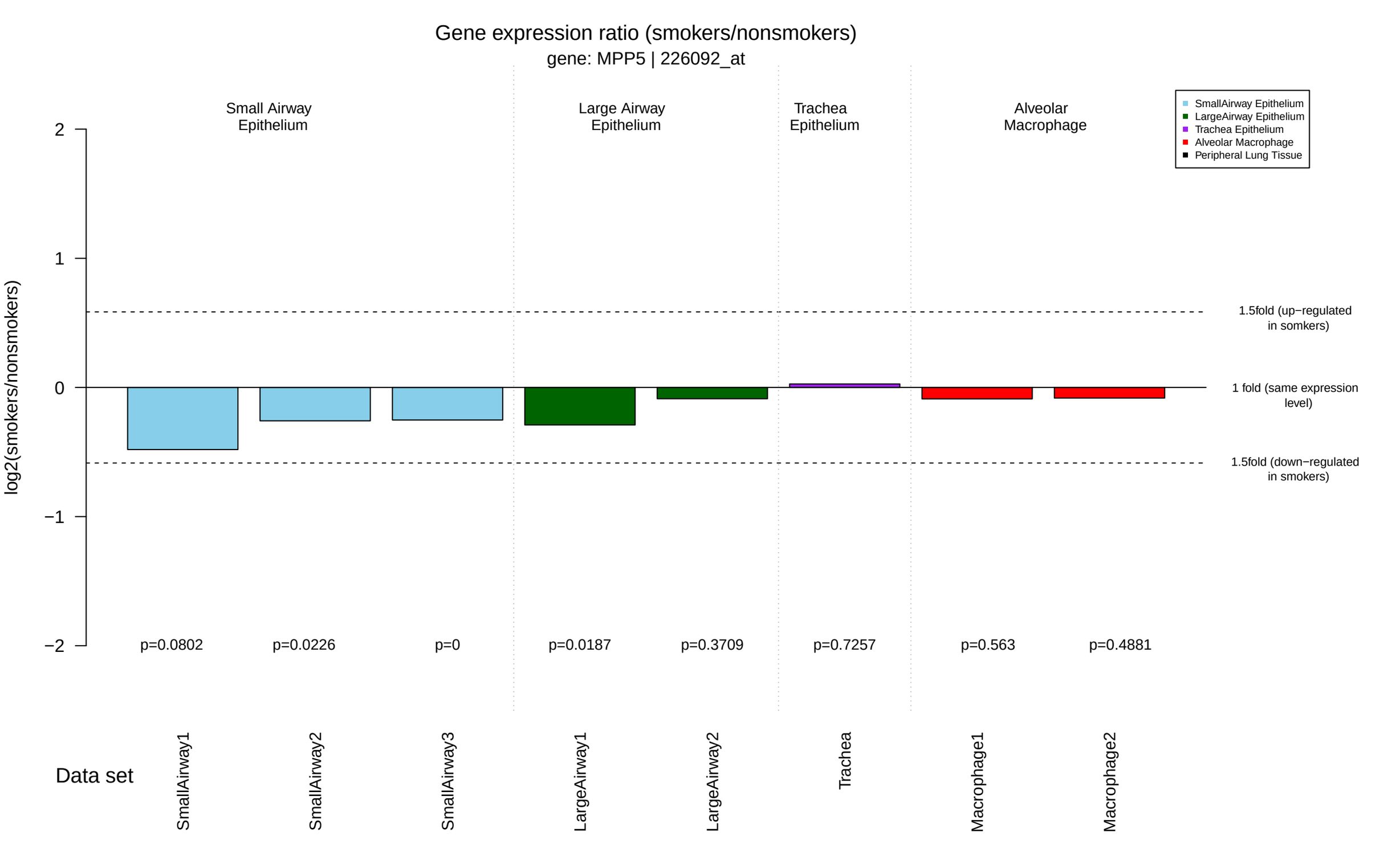
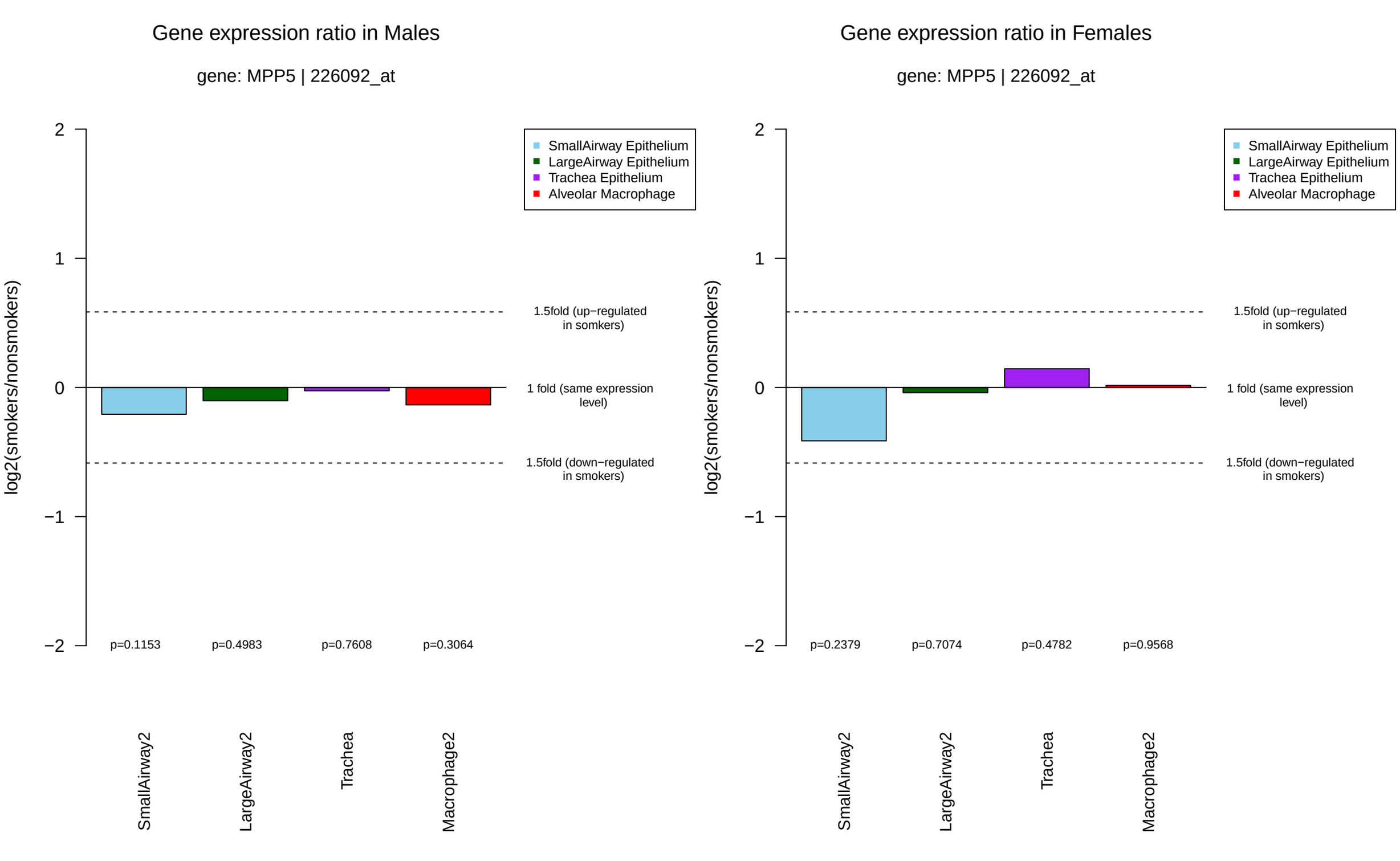
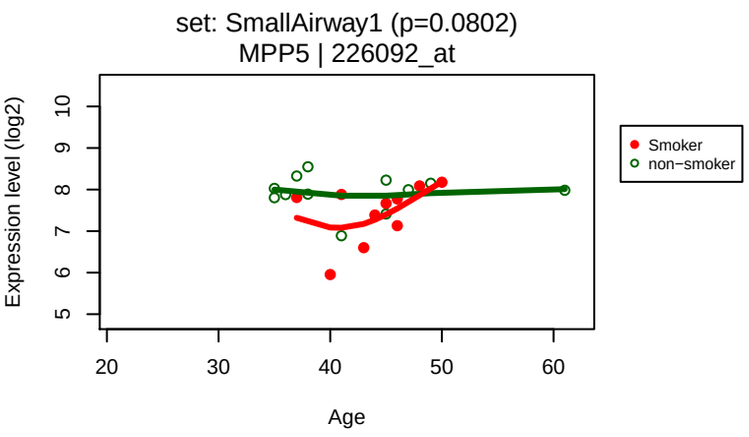
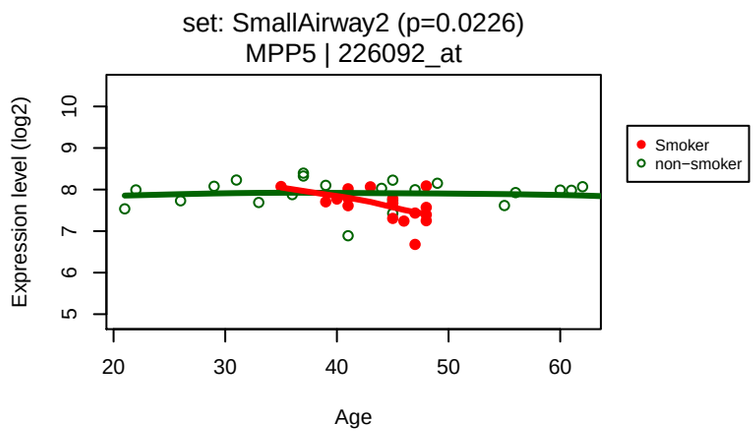
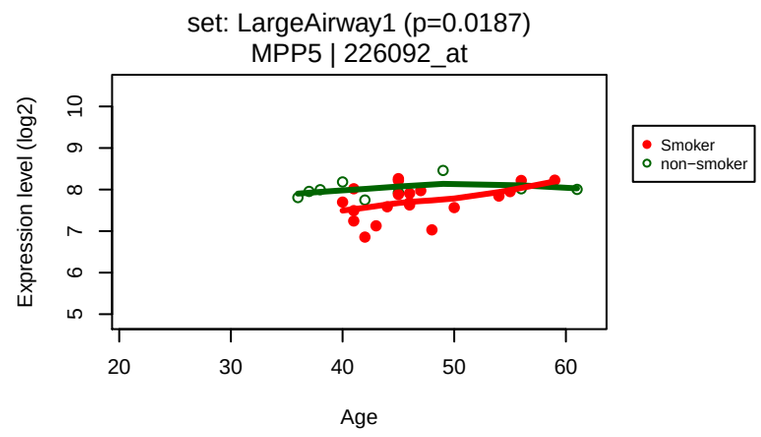

Gene expression of smokers and nonsmokers
gene: MPP5 | 226092_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0802
p=0.0226
p=0
p=0.0187
p=0.3709
p=0.7257
p=0.563
p=0.4881
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

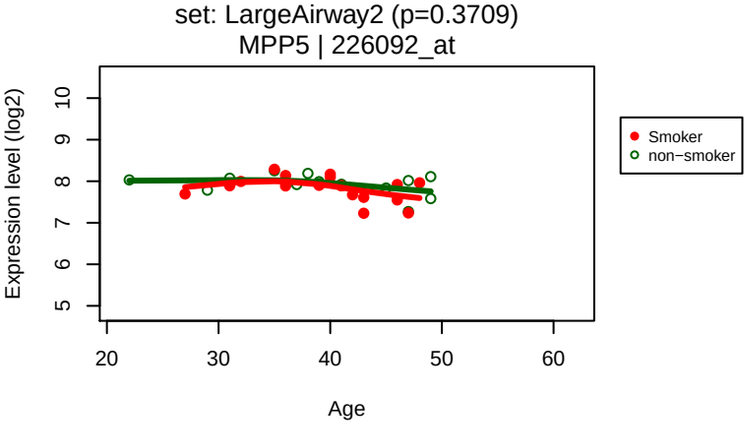

Gene expression ratio (smokers/nonsmokers)
gene: MPP5 | 226092_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0802
p=0.0226
p=0
p=0.0187
p=0.3709
p=0.7257
p=0.563
p=0.4881
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

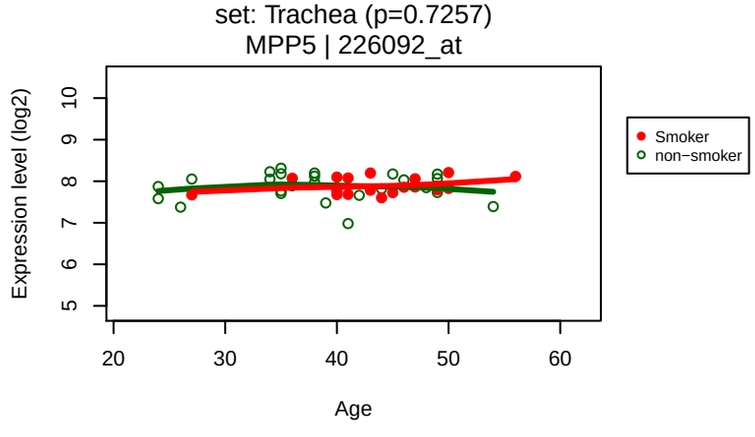

Gene expression ratio in Males
gene: MPP5 | 226092_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1153
p=0.4983
p=0.7608
p=0.3064
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MPP5 | 226092_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2379
p=0.7074
p=0.4782
p=0.9568
SmallAirway2
LargeAirway2
Trachea
Macrophage2

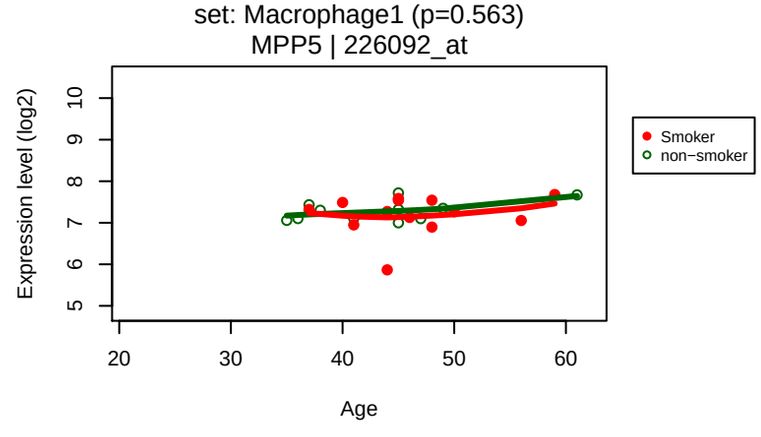

set: SmallAirway1 (p=0.0802)
MPP5 | 226092_at
Expression level (log2)
Age
Smoker
non−smoker

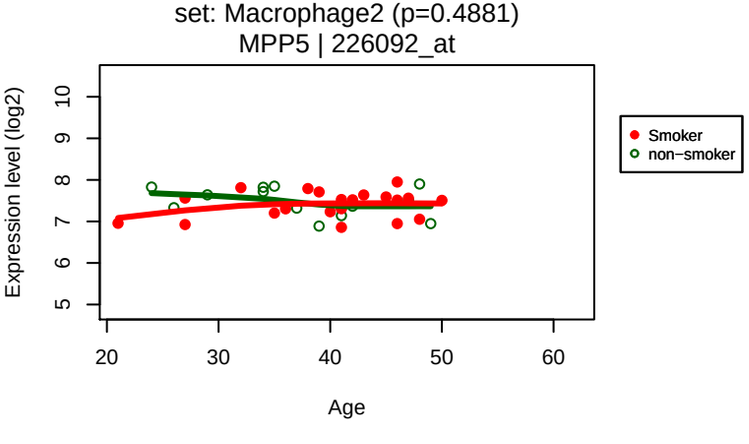

set: SmallAirway2 (p=0.0226)
MPP5 | 226092_at
Expression level (log2)
Age
Smoker
non−smoker

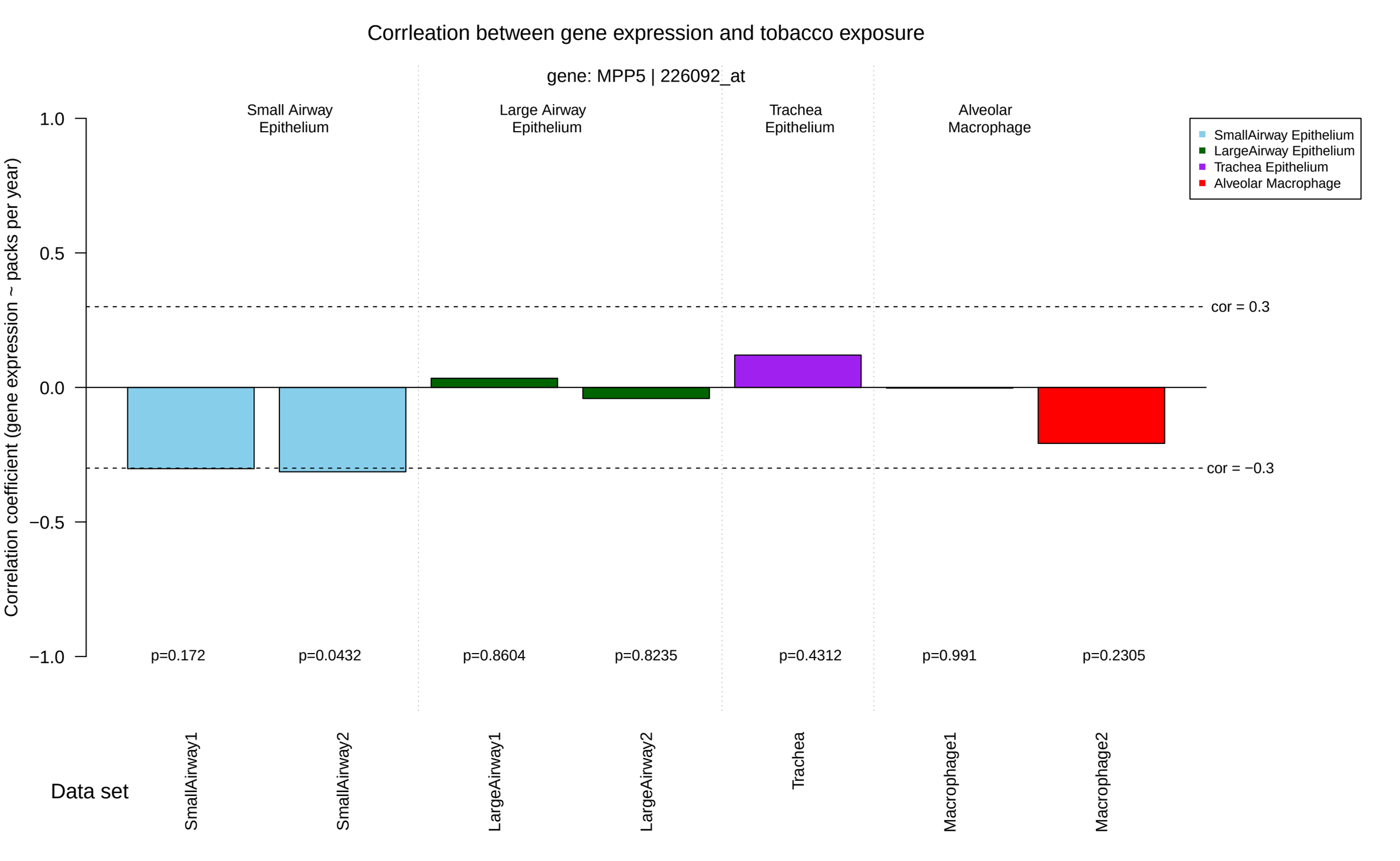

set: LargeAirway1 (p=0.0187)
MPP5 | 226092_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3709)
MPP5 | 226092_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7257)
MPP5 | 226092_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.563)
MPP5 | 226092_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4881)
MPP5 | 226092_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MPP5 | 226092_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.172
p=0.0432
p=0.8604
p=0.8235
p=0.4312
p=0.991
p=0.2305
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2